BridgeFinder
Finding Research Collaborators Across Communities at UIUC

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Abstract

Students who want research experience will often know an instructor in their area, but few researchers outside that instructor's circle. To address this, we made BridgeFinder, which helps them find related researchers and see who might connect them. We built a three-view prototype, analyzed UIUC publications from 2015 to 2024, and gathered feedback from ten students in two rounds. After initial testing, we found that BridgeFinder is better at predicting later collaborations than topic-only ranking, but not better than triadic closure. In the future, we plan on testing whether students find the suggestions relevant, unfamiliar, and useful for contacting others.

1 Objective

A student who majors in computer science might enjoy other topics like biology or agriculture, but not know of possible opportunities to apply their knowledge in an interdisciplinary way. UIUC lists 2,574 professor-titled faculty across 110 departments [15], far too many to browse. Searching for papers can reveal relevant names, but it does not tell the student whom they could approach or how to start a conversation.

BridgeFinder helps UIUC undergraduate and early-stage graduate students explore research beyond their existing contacts. A student chooses a familiar researcher, sees suggestions outside that person's collaboration community, and opens a card with topics, papers, and a possible introduction route. The starting researcher's work stands in for the student's interests, so the student needs no publications of their own.

This presents a difficult problem, as relevance, novelty, and reachability pull in different directions. For example, people who share a co-author with a familiar researcher might be easy to find and easy to reach, but they sit close to the familiar researcher in the co-authorship graph. People farther away are more likely to be new to the student, yet few of them work on related topics, and the student has no publications that could describe their interests. BridgeFinder must find the few distant researchers who are relevant and still reachable through people a student could ask.

For this midterm report, we have created a working three-view prototype, done a historical test on 49,651 UIUC papers that shows where new cross-community collaborations form, and taken in two rounds of student feedback, one that shaped four design changes and one that found no new problems.

2 Related Work

Researcher search. Illinois Experts [17] collects institutional profiles and publications, helping students check expertise. A profile helps assess a known researcher, while our task asks whom else to approach from a familiar person's network. Google Scholar [12] covers papers across fields and supports following topics through papers and author profiles. Those records still leave the student to work out a possible route of contact. BridgeFinder combines the research match with that route.

Referral chains. ReferralWeb [6] linked people whose names appear close together in Web documents and let users explore the resulting social network. A user could ask for their chain of acquaintances to a known researcher, or for colleagues and colleagues of colleagues who know a topic [9]. Showing the chain turns an expert into someone the user can reach. Chain and colleague queries start from the user's own place in the network, so the user must already appear in it, and the answers stay within a few links. BridgeFinder starts from a familiar researcher because students rarely appear in the graph, and it deliberately skips anyone within two hops.

Literature maps. Connected Papers [13] gives a visual overview of related papers, useful when entering an unfamiliar area. ResearchRabbit [14] supports continued exploration from paper collections. Both are built for literature discovery, and neither shows an introduction route from one researcher to another. We adapt their visual exploration to researchers and possible introductions.

Link prediction. Liben-Nowell and Kleinberg [7] compare ways to predict future links from network structure. Shared-contact counts provide a simple, explainable baseline, but favor people already nearby. LinkedIn's People You May Know [16] suggests members who share connections, employers, or schools with the user, so it also favors people near the user's network. We use the shared

co-author count as a simple triadic-closure baseline, not a reproduction of LinkedIn, to test whether our suggestions beyond shared contacts offer useful alternatives.

Cross-domain recommendation. Tang et al. [11] learn topics associated with cross-field collaboration and rank authors using random walks through author and topic nodes. This can capture complementary expertise, but relies on publication histories to represent authors. BridgeFinder serves students without papers by using a familiar researcher's topics as their interest proxy. It uses a fixed topic-and-path score and shows a possible introduction route. Our score does not learn relationships between topics across fields.

Network theory. Granovetter [5] explains how weak ties can carry information beyond close circles. Burt [3] explains the advantage of connecting separate groups. These ideas motivate looking beyond familiar collaborators. Neither establishes that an observed co-authorship route will lead to a useful introduction for a student.

3 Course Concepts and Network Analysis

3.1 Graph and communities

Brokerage involves connecting people across groups that have few ties to one another. We use this course concept to suggest researchers outside a familiar person's collaboration community. Nodes are UIUC authors from OpenAlex [10] papers. An edge joins two authors who wrote together, weighted by their number of joint papers.

On October 28, 2026, we downloaded all 102,947 OpenAlex works from 2015–2024 for UIUC (institution `I157725225`). We retained 49,651 with at least two distinct authors whose authorship records on that paper listed UIUC, and omitted external authors. Table 1 describes the 2015–2019 training graph.

Table 1. Training co-authorship graph from 2015–2019 papers.

Measure	Value
Authors	19,389
Edges	78,281
Communities	842
Modularity	0.884
Modularity in largest component	0.879

We use Louvain [2] (networkx, weighted, resolution 1, seed 42) to find communities with many collaborations inside each group. It seeks high *modularity*, meaning more within-group edge weight than expected from node strengths. Girvan–Newman [4] recomputes edge betweenness after every edge it removes, which is impractical for 78,281 edges, while Louvain partitions this graph in a few seconds.

Disconnected pieces can inflate modularity, but within the largest component, which holds 87% of authors, the same partition still scores 0.879. Collaboration therefore forms dense groups joined by few ties, the separation that brokers span. Communities are numbered groups, not verified departments.

3.2 Choosing suggestions

As in the proposal, candidates must be in another community and at least three hops from the starting researcher. One hop is one co-authorship edge. This excludes existing collaborators and people who already share a co-author. Triadic closure, the alternative covered in class, would suggest exactly those shared-contact people. It predicts ties well but keeps suggestions close to the familiar researcher, so we use it as a baseline instead. Candidates must also be reachable so we can show a route.

The ranking combines topic overlap with a bonus for introduction paths, as shown below.

$$\text{Score}(u, v) = S(u, v) + 0.5\frac{c_3(u, v)}{1 + c_3(u, v)}. \tag{1}$$

S is the number of shared topics divided by the number of distinct topics across both researchers, or zero if either topic set is empty. Each researcher has up to ten topics from their 2015–2019 papers. c_3 counts paths $u \to a \to b \to v$ with all four people distinct. Path counts and shortest routes ignore edge weights. The bonus stays below 0.5. Its saturation gives half of it to the first path, because one observed route already offers a possible introduction chain, and less to each extra route. More distant candidates remain eligible without that bonus. The coefficient and saturation are fixed choices, not learned probabilities.

For the 50 demo researchers, the path bonus dominates. All 500 suggestions are exactly three hops away, and the bonus makes up 68% of their summed scores, so the ranking favors people reachable through many short routes over the closest topic match. The 264 distinct intermediaries on these routes sit in broker-like positions. Their median Burt constraint [3], computed with edge weights on the training graph's largest component, is 0.12, against 0.47 for the median author there. This is consistent with collaborators who rarely work with one another, but constraint also falls as collaborators increase, so the comparison does not separate brokerage from connectivity.

Without the network, the app would lose its group boundary, distance rule, and introduction routes and become a topic search tool.

4 Design and Implementation

The prototype takes the student through three views.

1. Researchers. The student searches 50 starting researchers by name or topic. The proposal let students pick several instructors, but one starting researcher keeps each route readable. We chose highly connected authors across large communities so every choice has ten valid

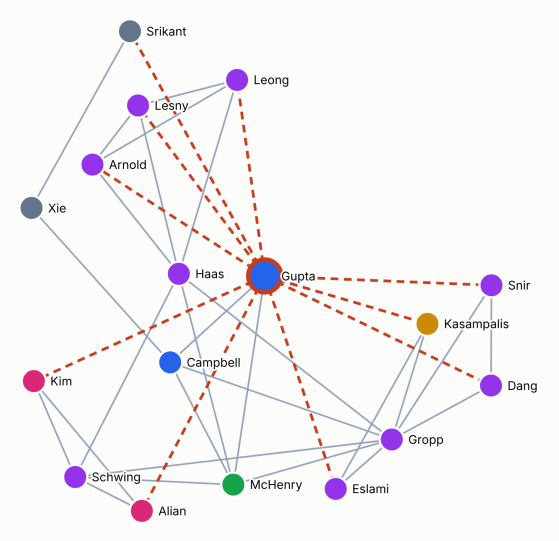


Figure 1. Network view for Indranil Gupta (ringed blue node). Dashed lines lead to the ten suggestions. Solid lines are observed co-authorships, including the intermediaries on each route. Colors mark communities.

suggestions, though a student's own instructor may be missing. As in profile search [17], it provides an entry point without requiring a student profile.

2. Network. A map shows the selected researcher, suggestions, and intermediaries, colored by community (Figure 1). Solid lines show observed co-authorships, and dashed lines show suggested connections. This borrows visual exploration from literature maps [13, 14] while focusing on people. A ranked list of the ten suggestions sits beside the map, merging the proposal's map and list screens. The proposal planned a campus-wide map, but feedback led us to show only this neighborhood (Section 5).

3. Suggestions. Selecting a name in the list or a dashed line opens a card with an observed shortest route, topics, and example papers. We put the route first to show whom a student might approach, following ReferralWeb's referral chains [6]. An optional explanation shows the ranking score. Students can edit, copy, or download an outreach draft. Figure 2 shows a card.

Because the path bonus dominates the score, a suggestion can have little topic overlap. The topics and papers below the route let students judge relevance themselves, and the study's relevance ratings will show whether the bonus should be reduced.

The app, historical test, and study cards share one ranking implementation. Recommendations are computed in Python and displayed in D3. The core flow works on desktop and mobile and has been browser-tested for all 50 starting researchers.

5 Users and Feedback

Our target users want research opportunities but have few contacts beyond their classes or department. We gathered feedback from ten UIUC students in two rounds. In the first round, the two graduate students were a Computer Science student and a Master of Computer Science student researching computational biology. One undergraduate studied Electrical and Computer Engineering, and the other studied Food Systems.

We gave these students minimal context in individual sessions and asked them to work through the app, and they gave feedback as they encountered problems. Together they cover both levels we target. The Food Systems student represents the non-computing side of the cross-field searches BridgeFinder targets, and the computational biology student has already made such a cross-field move. This round had four testers, below the five to ten the proposal cited, and the second round added six. Small formative rounds can expose major problems before a broader study [8], but they cannot estimate usability across campus. We made four changes after the first round.

Getting started. One student trying to find a relevant professor did not understand that they first needed to select a researcher to enter the graph. We added instructions on the landing page explaining that students should choose a starting researcher, explore a suggestion, and then review its route.

Understanding the interface. Students found the interface unintuitive, prompting an overhaul with simpler layouts and three views named Researchers, Network, and Suggestions. They were also unsure what a card meant. The landing page now defines a researcher card as a profile with topics, example papers, and a possible introduction route. The map legend distinguishes observed collaborations from suggestions.

Readable map. The graph showed too many nodes and was difficult to follow. We narrowed the display to the selected researcher, suggestions, and people along their routes.

Correct labels. Feedback also exposed tags in the original design that did not match the researchers shown. We corrected the profile labeling and now distinguish inferred research areas from collaboration communities.

Second round. Six more UIUC students then reviewed the revised app. Two were undergraduates studying mathematics and computer science, one in a dual degree and one in the combined degree. Three were Master of Computer Science students, and one was a master's student in business technology management. All six found the revised interface clear, raised no new problems, and approved of the app. Most testers so far study computing, so the study in Section 6.2 will test it with 15 students from a wider range of fields.

6 Evaluation

6.1 Historical prediction

We used the papers published between 2015 and 2019 from OpenAlex to build the graph, communities, and

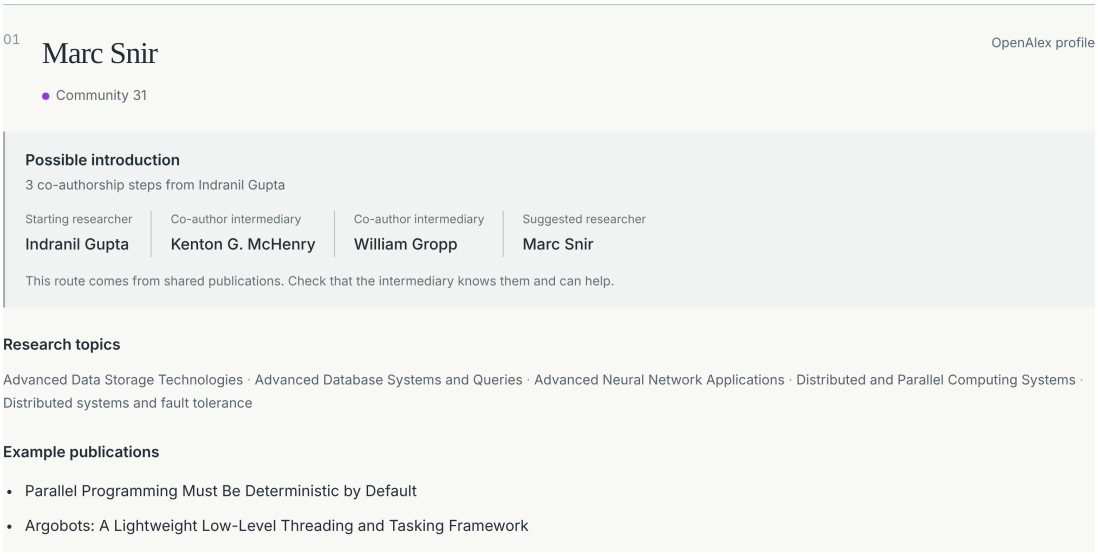


Figure 2. Card for Marc Snir, the top suggestion for Indranil Gupta. The route runs from Indranil Gupta through Kenton G. McHenry and William Gropp. Topics and example papers below it let a student judge relevance. The route is observed in publications, not a confirmed introduction.

topic profiles. We then checked for new cross-community co-authorships in papers from 2020–2024. We evaluated 3,492 starting researchers with at least one new cross-community tie to another training-graph author. That tie could be two hops away, more distant, or disconnected in training.

All methods rank training-graph authors in another community and exclude existing collaborators. **Topic-Only** ranks BridgeFinder's reachable candidate pool at distance three or more by S alone. **Triadic closure** ranks two-hop candidates by their number of shared co-authors, without additional weighting. All methods predict the same future ties, so the comparison is between discovery strategies with different candidate pools.

Precision at ten, or **P@10**, is the share of each researcher's first ten suggestions who became new cross-community collaborators in the later period. We divide hits by ten even when fewer suggestions are available, then average across starting researchers.

Table 2. Prediction of new 2020–2024 cross-community ties. Higher is better.

Method	P@10
Topic-Only	0.0079
Triadic closure	0.0287
BridgeFinder	0.0131

Table 2 shows that BridgeFinder predicts more later ties than Topic-Only (0.0131 vs 0.0079), but fewer than triadic closure (0.0287). It therefore does not meet the proposal's goal of outperforming both baselines. Shared contacts still miss most new ties. Of the 8,916 distinct new cross-community pairs behind Table 2, only 22% joined authors who already shared a co-author, while 67% joined authors three or more hops apart, inside BridgeFinder's candidate pool, and the remaining 11% had no path in the training graph. These shares describe where new ties formed, not how likely any one distant pair is to collaborate, because distant candidates far outnumber two-hop ones. Because the pools differ, the triadic-closure comparison cannot separate the effect of distance from the effect of the score. The student study will test whether the suggestions are useful for discovery despite lower historical prediction than triadic closure.

6.2 Planned student study

We will recruit 15 undergraduate and early-stage graduate students from a wide variety of backgrounds who want research experience.

Each student will first state a research interest, then choose a starting researcher, open a suggestion and outreach draft, and explain the route and dashed line without coaching. We will record task completion and whether the student identifies the intermediaries and distinguishes suggestions from existing ties.

For the card comparison, students will choose the closest match to their interest from five starting researchers and rate up to 15 shuffled cards, five per method. Method labels are hidden. A candidate suggested by two methods appears once and counts for both. All cards use the same fields and shortest-path procedure, with topics in place of the proposal's departments, which OpenAlex does not record reliably. Triadic-closure cards show two-hop routes, while BridgeFinder and Topic-Only show routes of three or more hops, so route length can reveal triadic-closure cards and blinding is partial for that comparison. We will collect the following measures.

- **Relevant discovery.** Students will rate relevance to their stated interest and novelty to themselves on scales from 1 to 7, a finer version of the proposal's yes-or-no familiarity question. Higher ratings mean a better research match or a less familiar suggestion. For BridgeFinder, we target means of at least 5 on both scales, and a share of cards rated at least 5 on both that is 25 percentage points above triadic closure, our primary comparison. Its share should also exceed Topic-Only's.
- **Willingness to contact.** Students will rate how realistic contact feels given the route on a scale from 1 to 7. This checks whether they see a practical next step. For BridgeFinder, we target a mean of at least 5 and improvement over Topic-Only. Topic-Only draws on the same candidates without the path bonus, so this comparison asks whether favoring well-connected candidates makes contact feel more realistic. Triadic closure suggests people who share a co-author, so we report it for reference without a target.
- **Usability.** At least 12 of 15 students should complete the task and explain the line types correctly without help. The ten-item System Usability Scale will assess ease of using the interactive app. We target a mean of 80/100, above typical scores in published SUS norms [1].

We will compare methods by participant averages rather than treating individual cards as independent responses. The rating and analysis tools are implemented. We have not yet run the study.

7 Limitations and Next Steps

Even though we can identify a pathway through co-authorship, we cannot verify that the intermediaries would introduce a student, or that the suggested researcher is available or has the resources to mentor one. Our dataset also misses ties to authors outside UIUC and informal ties that never produced a paper, and OpenAlex does not capture every publication. The starting researcher's topics may not reflect a student's own interests. OpenAlex affiliations and dates also add noise. Of the 500 displayed suggestions, 29% appear on only one UIUC paper from 2015–2024, and one affiliated paper does not establish a current affiliation with UIUC.

Two thirds of new cross-community collaborations joined people three or more hops apart, so looking past shared contacts captures where most of them formed, even though any single distant pair is less likely to collaborate. Triadic closure still predicts these new connections better, while BridgeFinder beats Topic-Only ranking by also using network position.

In November, we will run the student study in Section 6.2 to test whether students find the suggestions relevant, new, and worth acting on. Any ranking changes, such as requiring at least two UIUC papers per suggestion, will use separate validation data.

8 AI Use

We used Claude for early development of the prototype and to draft an early version of the paper. We also used Perplexity and OpenAI Codex to check the work and revise it. We read and verified everything ourselves.

References

[1] A. Bangor, P. T. Kortum, and J. T. Miller. An empirical evaluation of the System Usability Scale. *International Journal of Human–Computer Interaction*, 24(6):574–594, 2008.

[2] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008.

[3] R. S. Burt. *Structural Holes: The Social Structure of Competition*. Harvard University Press, Cambridge, MA, 1992.

[4] M. Girvan and M. E. J. Newman. Community structure in social and biological networks. *Proceedings of the National Academy of Sciences*, 99(12):7821–7826, 2002.

[5] M. S. Granovetter. The strength of weak ties. *American Journal of Sociology*, 78(6):1360–1380, 1973.

[6] H. Kautz, B. Selman, and M. Shah. Referral Web: Combining social networks and collaborative filtering. *Communications of the ACM*, 40(3):63–65, 1997.

[7] D. Liben-Nowell and J. Kleinberg. The link-prediction problem for social networks. *Journal of the American Society for Information Science and Technology*, 58(7):1019–1031, 2007.

[8] J. Nielsen. *Usability Engineering*. Morgan Kaufmann, San Francisco, CA, 1993.

[9] S. Perugini, M. A. Gonçalves, and E. A. Fox. A connection-centric survey of recommender systems research. *arXiv preprint cs/0205059v2*, 2003.

[10] J. Priem, H. Piwowar, and R. Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*, 2022.

[11] J. Tang, S. Wu, J. Sun, and H. Su. Cross-domain collaboration recommendation. In *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '12)*, pages 1285–1293, 2012.

[12] Google. Google Scholar: About. n.d. `https://scholar.google.com/intl/en/scholar/about.html`.

[13] Connected Papers. Connected Papers: Explore connected papers in a visual graph. n.d. `https://www.connectedpapers.com/`.

[14] ResearchRabbit. ResearchRabbit: Literature discovery. n.d. `https://www.researchrabbit.ai/`.

[15] University of Illinois Board of Trustees. Academic and administrative appointments (Gray Book), February 2024. `https://bot.uillinois.edu/resources/gray_book/`.

[16] LinkedIn. People You May Know feature. LinkedIn Help. n.d. `https://www.linkedin.com/help/linkedin/answer/29`.

[17] University of Illinois. Illinois Experts: Research and scholarship at UIUC. n.d. `https://experts.illinois.edu/`.